

LESSON 1.3

APPLYING THE LAWS OF EXPONENTS – PART 2



LESSON INTRODUCTION

MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

LEARNING TARGETS

- I can use the quotient of powers, power of a quotient, and zero exponent laws to simplify expressions.
- I can rewrite expressions using the laws of exponents to generate equivalent expressions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.

WORKING TOWARD

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

ESSENTIAL QUESTIONS

- What are the Laws of Exponents for division?
- How can you use the Laws of Exponents to simplify and create equivalent expressions?

LESSON NARRATIVE

Students build on their knowledge of the Laws of Exponents for products to begin exploring the Laws of Exponents for quotients. Throughout the lesson, students will work with the Quotient of Powers, Power of a Quotient, and Zero Exponent Laws. The lesson concludes with students using the exponent laws to write and analyze equivalent exponential expressions.

COMPONENT SUMMARY

1.3.1 WARM-UP (~5 MINS)

Notice and Wonder about a bridge

1.3.3 BRING IT TOGETHER (5 MINS)

Students are introduced to the formal definitions for the Laws of Exponents

1.3.5 COLLABORATION (5-10 MINS)

Students collaborate to analyze equivalent expressions

1.3.2 EXPLORATION (10-15 MINS)

Students discover the Laws of Exponents by doing examples

1.3.4 YOUR TURN! (10-15 MINS)

Students practice rewriting expressions and identify the law(s) used

1.3.6 WRAP-UP (~5 MINS)

Emoji self-reflection activity

RESOURCES

GLOSSARY

Key Terms:

- Quotient of Powers Exponent Law
- Power of a Quotient Exponent Law
- Zero Exponent Law

Additional Terms:

- N/A

MATERIALS

- (Optional) Index cards

ADDITIONAL PREPARATION

- (Optional) 1.3.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe that a negative sign belongs only with the numerator or only with the denominator in a fraction.
- Students may believe that any number with an exponent of zero equals zero.

THINGS TO CONSIDER

- Help students understand that fractions represent division. Thus, either a negative numerator or a negative denominator would make the entire fraction negative.
- Students can explore the Zero Exponent Law to understand why an exponent of zero does not equal one.

LITERACY AND LANGUAGE ROUTINES

Co-Craft Questions and Problems

The Exploration invites students to use mathematical patterns and expanded form to deduce the Laws of Exponents for quotients. Before students begin the activity, encourage them to broadly think about their observations. Then, ask students to narrow down the details as they hone their conclusions and knowledge about the mathematical processes at play.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students will be working to simplify and rewrite expressions using the Laws of Exponents for quotients. Clarifications within this standard emphasize accurately performing mathematical procedures and calculations. Utilize the progression of the lesson to allow students to practice efficient and generalizable mathematical methods.

DIFFERENTIATE INSTRUCTION

Allow students to respond in multiple ways including written, verbal, and by using examples. Set up stations around the room for students to move through as they complete a series of problems. Consider transforming some of the tables into matching activities for students to work on in small groups.

Additional differentiation opportunities are denoted throughout the lesson guide.

1.3.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Encourage students to think about many different aspects of the structure, such as what it is made of, how it may have been created, and how much it weighs.

QUESTIONING

Share M.C. Escher's Ascending and Descending from with students. Ask students how the structures are similar and how they are different. Encourage students to consider what the internal, unseen structures of each would be if the buildings were real.



1.3.1 Warm-Up: Notice and Wonder

In May 2021, the world's longest suspended pedestrian bridge in the world opened in Arouca, Portugal. The bridge runs over 1700 feet across the canyon, nearly 600 feet above the Paiva River.



1. What do you notice?

Answers will vary. I notice that there are a lot of geometric figures used to create the bridge, including hexagons and triangles. I notice the bridge sags in the middle.

2. What do you wonder?

Answers will vary. I wonder if the bridge sways back and forth in the wind. I wonder how long it took to build the bridge. I wonder how they were able to construct the bridge over the canyon.



1.3.2 Exploration: Exponent Laws

1. Evaluate each of the following expressions.

$$\frac{10}{10} = 1$$

$$\frac{-7}{-7} = 1$$

$$\frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

2. If the numerator and denominator have the same value, the quotient is 1.
3. The **Quotient of Powers Exponent Law** can be used to rewrite expressions when dividing two powers that have the same nonzero base.
- a. Complete the table.

Quotient	Expanded form	Exponential Form a^b
$\frac{5^5}{5^2}$	$\frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5}$	5^3
$\frac{(-4)^7}{(-4)^6}$	$\frac{-4 \cdot -4 \cdot -4 \cdot -4 \cdot -4 \cdot -4 \cdot -4}{-4 \cdot -4 \cdot -4 \cdot -4 \cdot -4 \cdot -4}$	-4
$\frac{(\frac{3}{4})^6}{(\frac{3}{4})}$	$\frac{(\frac{3}{4}) \cdot (\frac{3}{4}) \cdot (\frac{3}{4}) \cdot (\frac{3}{4}) \cdot (\frac{3}{4}) \cdot (\frac{3}{4})}{(\frac{3}{4})}$	$(\frac{3}{4})^5$

- b. How do the exponents in the first column relate to the exponents in the third column?
The exponents in the first column relate to the exponents in the third column because subtracting the exponents in the first column will result in the exponent in the third column.

- c. Work with your partner to complete the rule for finding the quotient of powers.

$$\frac{a^m}{a^n} = \begin{cases} a^{mn} \\ a^{m-n} \\ a^{m+n} \end{cases}$$

4. The **Power of a Quotient Exponent Law** can be used to rewrite expressions when a fraction is raised to an exponent.

- a. Complete the table.

Expression	Expanded Form	Exponential Form a^b
$(\frac{2}{5})^4$	$\frac{2}{5} \cdot \frac{2}{5} \cdot \frac{2}{5} \cdot \frac{2}{5}$	$\frac{2^4}{5^4}$
$(-\frac{4}{7})^3$	$-\frac{4}{7} \cdot -\frac{4}{7} \cdot -\frac{4}{7}$	$-\frac{4^3}{7^3}$

- b. How do the exponents in the first column relate to the exponents in the third column?
The exponents in the first column relate to the exponents in the third column because the exponent in the first column must be applied to both the numerator and denominator in the third column.

1.3.2 EXPLORATION: EXPONENT LAWS

TEACHER MOVES

Co-Craft Questions and Problems

Have students create their own Quotient of Powers problems and discuss with a partner why they set up their problems the way they did. Use this routine to allow students to create meta-awareness of the language and process used in the Laws of Exponents for quotients.

ENRICHMENT

Why does a fraction with the same numerator and denominator equal 1? How does this relate to a fraction as a form of division? What pattern do you notice about fractions with exponents?

TEACHER MOVES

Have students discuss the position of the negative sign in a fraction and what fractions represent. Draw students' attention to the negative sign placed in the middle of the fraction in question 4a.

ENRICHMENT

What pattern do you notice about raising a fraction to an exponent? How can the Power of a Quotient Exponent Law help you remove the parentheses in an expression? Does a similar law apply to whole numbers? Why or why not?

1.3.2 EXPLORATION: EXPONENT LAWS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider turning the table into a matching activity. Write the column headings and the answers for each section on individual index cards. Shuffle the cards for one table and give them to a small group of students. Have students work together to discuss and analyze the cards as they work to create the table. As students work, circulate throughout the room to ask them to explain the choices they make.

QUESTIONING

Offer students some more challenging expressions to solve, such as ones that combine operations with exponential numbers.

- c. Work with your partner to complete the rule for finding the power of quotients.

$$\left(\frac{a}{b}\right)^m = \begin{cases} \circ a^b - m, \\ \circ \frac{a^m}{b^m}, \\ \circ a^m - b^m. \end{cases}$$

5. The **Zero Exponent Law** is used to rewrite expressions when a nonzero rational number is raised to the zero power.

- a. Complete the table.

Quotient	Quotient of Powers Law	Expanded Form	Simplified Form
$\frac{4^5}{4^5}$	$4^{5-5} = 4^{\boxed{0}}$	$\frac{\overbrace{4 \cdot 4 \cdot 4 \cdot 4 \cdot 4}^5}{\underbrace{4 \cdot 4 \cdot 4 \cdot 4 \cdot 4}_5}$	1
$\frac{(-5.7)^3}{(-5.7)^3}$	$\frac{-5.7^{3-3}}{-5.7^0} = (-5.7)^0$	$\frac{\overbrace{-5.7 \cdot -5.7 \cdot -5.7}^3}{\underbrace{-5.7 \cdot -5.7 \cdot -5.7}_3}$	1
$\frac{(\frac{1}{2})^6}{(\frac{1}{2})^6}$	$\frac{(\frac{1}{2})^{6-6}}{(\frac{1}{2})^0} = (\frac{1}{2})^0$	$\frac{\overbrace{(\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2})}^6}{\underbrace{(\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) \cdot (\frac{1}{2})}_6}$	1

- b. Complete the statement.

Any nonzero rational number raised to the 0 power is equal to 1.



1.3.3 Bring It Together: Exponent Laws



Quotient of Powers Exponent Law

$$\frac{a^m}{a^n} = a^{m-n} \text{ and } a^{m-n} = \frac{a^m}{a^n}$$



Power of a Quotient Exponent Law

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \text{ and } \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$



Zero Exponent Law

$$a^0 = 1$$

1.3.3 BRING IT TOGETHER: EXPONENT LAWS

TEACHER MOVES

Some students may struggle with the Zero Exponent Law. Consider using the quotient rule with any non-zero rational number to model the Zero Exponent Law. Since any number (other than zero) divided by itself is equal to one:

$$\frac{2^5}{2^5} = 2^{5-5} = 2^0$$

And since $\frac{2^5}{2^5} = 1$
then $2^0 = 1$

ENRICHMENT

- How could you write a problem to show the Quotient of Powers Exponent Law?
- How would you explain the Power of a Quotient Exponent Law in your own words?
- Why does any number raised to the zero power equal one?

1.3.4 YOUR TURN!

TEACHER MOVES

MA.K12.MTR.3.1: Mathematical Fluency

Encourage students to select the appropriate exponent law for rewriting each expression. If it helps students better understand the process, have them explain their reasoning and process to a partner.

QUESTIONING

Assign students an exponent law. Then have them create an exponential expression and an equivalent expression to represent it. Repeat with additional Laws of Exponents as time allows.

ENRICHMENT

- How are the Power of a Product and the Power of a Quotient Laws similar?
- What do they each allow you to do?
- What do the Product of Powers and Quotient of Powers Laws have in common?
- How does knowing one of the laws help you remember the other?

DIFFERENTIATE INSTRUCTION

Set up seven stations around the classroom with one row of problems at each station. When students correctly complete a problem, direct them to the next station. This process gets students moving and helps keeps them engaged with the activity. Students may complete the activity in small groups and discuss each problem with their partners.



1.3.4 Your Turn!

1. Complete the table by rewriting each expression as an equivalent expression using the laws of exponents.

Exponential Expression	Equivalent Expression	Exponent Law(s) Applied
$\frac{(1.3)^6}{(1.3)^3}$	$(1.3)^3$ 2.197	☐ Product of Powers ☒ Quotient of Powers ☐ Power of a Product ☐ Power of a Quotient ☐ Power of a Power ☐ Zero Exponent
$\left(\frac{3}{-5}\right)^5$	$\frac{3^5}{(-5)^5}$ $\frac{243}{-3125}$ -0.07776	☐ Product of Powers ☐ Quotient of Powers ☐ Power of a Product ☒ Power of a Quotient ☐ Power of a Power ☐ Zero Exponent
$(-3)^2 \cdot (-3)$	$(-3)^3$ -27	☒ Product of Powers ☐ Quotient of Powers ☐ Power of a Product ☐ Power of a Quotient ☐ Power of a Power ☐ Zero Exponent
$(-4)^0$	1	☐ Product of Powers ☐ Quotient of Powers ☐ Power of a Product ☐ Power of a Quotient ☐ Power of a Power ☒ Zero Exponent

$\left(-\frac{3}{15}\right)^{15}$	$\frac{(-3)^{15}}{15^{15}}$	☐ Product of Powers ☐ Quotient of Powers ☐ Power of a Product ☒ Power of a Quotient ☐ Power of a Power ☐ Zero Exponent
$\frac{(-7.45)^{11}}{(-7.45)^{11}}$	$(-7.45)^0$ 1	☐ Product of Powers ☒ Quotient of Powers ☐ Power of a Product ☐ Power of a Quotient ☐ Power of a Power ☒ Zero Exponent
$\frac{9^9}{9^4}$	9^5 59,049	☐ Product of Powers ☒ Quotient of Powers ☐ Power of a Product ☐ Power of a Quotient ☐ Power of a Power ☐ Zero Exponent



1.3.5 Collaboration: Analyzing Equivalent Expressions

Noah and Elena each generated a list of equivalent expressions by applying exponents laws.

Work with your partner to analyze each list for errors.

- Noah generated the following expressions equivalent to 5^9 .

5^{46}	$(5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5)$	$(\frac{-10}{-2})^9$
$\frac{5^{12}}{5^3}$	$5^7 \cdot 5^2$	$(\frac{5}{-5})^9$

Circle the equivalent expressions and draw an X through the expressions that are not equivalent to the original expression.

- Elena generated the following expressions as equivalent to $(2.95)^0$.

1	0	10^0
$\frac{2.95^0}{2.95^0}$	2.95	$(\frac{1}{10})^{17}$

Circle the equivalent expressions and draw an X through the expressions that are not equivalent to the original expression.



1.3.6 Wrap-Up: Reflection

Circle the value on the scale that matches your understanding of each learning target.

Answers will vary.

- I can use the quotient of powers, power of a quotient, and zero exponent laws to simplify expressions.



- I can rewrite expressions using the laws of exponents to generate equivalent expressions.



1.3.5 COLLABORATION: ANALYZING EQUIVALENT EXPRESSIONS

TEACHER MOVES

Consider using flexible groups as needed to help students maintain interest in the topic. Some students may prefer to work independently, while others will do well discussing the problems with a partner. Also, consider forming groups of six students with each student determining if one of the expressions is equivalent to the given expression, and then explaining their answer and reasoning to the other members of the group.

ENRICHMENT

Why are the expressions not equal to 59? What are they equal to? How did you determine which expressions are equal to $(2.95)^0$?

1.3.6 WRAP-UP: REFLECTION

TEACHER MOVES

Encourage students to respond in as much detail as possible regarding what they do and do not understand in the lesson. Use students' responses to plan future instruction. Form a small group for students who could benefit from reteaching. Also, consider which students may be able to progress more quickly through future lessons.

DIFFERENTIATE INSTRUCTION

Allow students to respond in multiple ways. Some students may prefer to explain their reasoning in writing, while others may prefer to discuss it orally. Some students may benefit from using an example to show what they do and do not understand about working with the Laws of Exponents for quotients.

QUESTIONING

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.